



ZIGMA GLOBAL

A Pioneer in Landfill Waste Management



THE PROBLEM IS MULTIFOLD:

1. *The waste is unsegregated and the Urban or Local Bodies (ULB's) do not have infrastructure to segregate them.*
2. *The waste is forcibly dumped in open dumpsites as a result.*
3. *The waste ends up in landfills, giving no scope to the Municipal Bodies to build new infrastructure, and also contaminating soil and underground water, polluting air, etc.*
4. *The neighbours protest when the ULB's try to acquire new land for landfills.*

As India heads to become the world's largest populated country, the problem of haphazard dumping of MSW is increasing, causing irreparable damage to public health and the environment. Integrated Landfill Mining ensures that the haphazardly dumped MSW in open landfills is mined and completely removed, addressing the problems of air, water pollution and soil contamination and helping provide a clean environment to the poor people living nearby these landfills and the society at large.

COMPANY OVERVIEW

Incepted in 2015, Zigma Global Environ Solutions is a pioneer and currently the largest integrated landfill mining company in India. The company has till today reclaimed more than a hundred and fifty acres of land from landfills, and processed, segregated and responsibly disposed of more than 12 lakh metric tonnes of legacy waste from various landfill sites.

SOME OF THE KEY FEATURES OF THE SERVICES PROVIDED BY ZIGMA INCLUDE:

- *Zigma provides an integrated service in Design Build and Operate model wherein the ULB's need not invest in plant and machineries or any capital expenditure. Moreover, the landfill site is handed over to the company and once the mining is completed, the company gives back the*

reclaimed site to the ULB.

- *The machines used are completely mobile, and can be completely dismantled and removed on completion of the project.*
- *The model is scalable depending on the quantity of waste present in the dumpsite.*
- *The company offers a near zero residue principle of waste management and hence 100% land is reclaimed, leaving no traces of further contamination or pollution.*
- *The service provided completely adheres to SWM Rules 2016 and CPCB Guidelines for disposal of legacy waste 2019.*

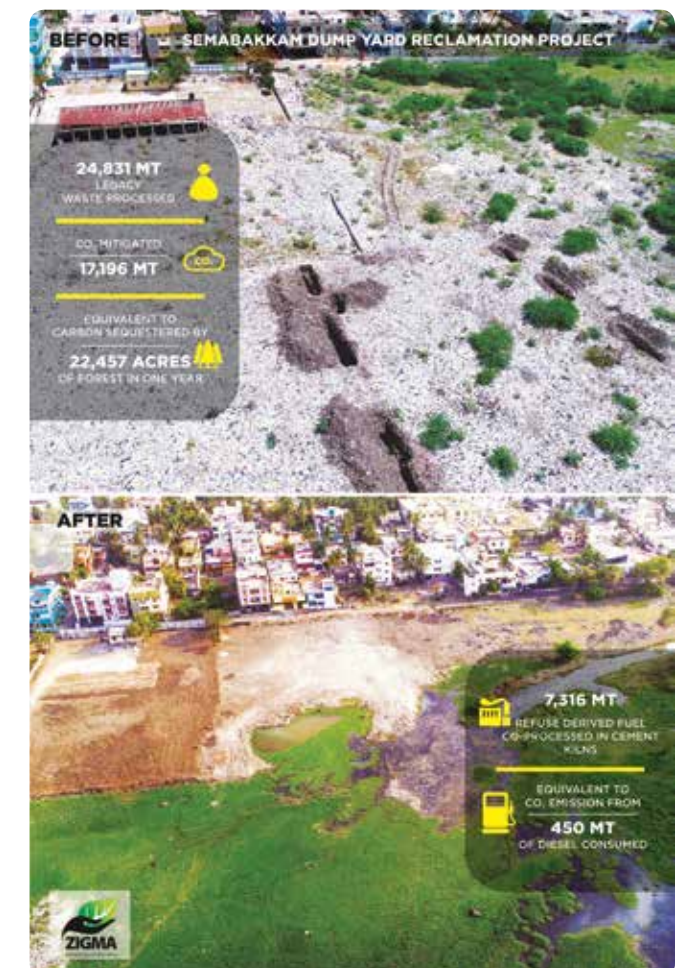
The company was founded five years ago by like-minded four serial entrepreneurs B. Dharmaraj, K.P. Mutharasu, K.T. Ilangovan and Nagesh Prabhu C., who are in the field of steel manufacturing, UPVC extrusion and IT Infrastructure. The idea was to start a business platform in waste management in a space which was not explored professionally in the Indian subcontinent. The Co-founders travelled to South East Asia, Middle East and Europe to understand the waste management methodologies adopted by these countries and how relevant

they could be if adopted in India. A similar research was conducted in India to understand the lacunas and the area identified was legacy waste management, which was the biggest challenge that India with a population of 1.3 billion people faced.

The co-founders then decided not to resort to off-the-shelf solutions, and identified key persons from various industries to self-design, build and commission machines for handling the legacy waste with a vision to segregate the legacy waste and recycle, reuse or repurpose all the aggregates that were generated.

Creating a team of committed individuals is an important part in establishing any enterprise. Since the Co-founders were already from the manufacturing sector, they identified the key persons from various industries who had vast experience in designing and manufacturing machines. All the machines were designed from the drawing boards to customize them to achieve efficiencies which was unheard of with a target to achieve near zero residues. This team became the backbone of the company which later on recruited others who were best in the industry and the vision was passed on down the line.

Under their leadership and with a team committed to the cause, Zigma Global has been an indispensable aid in managing the legacy waste concern in India.



Urban India generates about 1.5 lakh MT of Municipal Solid Waste (MSW) daily, out of which less than 10% is processed and all the rest is unscientifically dumped. There are about 4,500 identified dumpsites in urban India alone and many more remain yet to be identified. The Solid Waste Management Rules 2016 mandate reclamation of all these dumpsites with technologies that leave little or no residues. The country has been grappling with the challenge in proper management of MSW landfills from the days plastic was introduced in households for various applications.



waste. It has mastered the art of pre-stabilisation of waste which is the most and integral part of the landfill mining process without which near zero residues concept cannot be implemented. Zigma has built a state-of-the-art internal and external ERP system which allows SOP's in all areas of operation. The company has associations with renowned research institutes across the country which provide consultancy on various technical parameters.

The company has been a pioneer in providing integrated solutions. While other companies in the space provide piece meal solutions, Zigma provides an end to end solution from pre-stabilization to disposal of all aggregates. It provides solutions on a Design Build Own and Operate model where the ULB's don't have to invest on the capital for buying machines, as these machines once work is completed are of no use for the ULB's to retain.

Zigma Global Environ Solutions Pvt Ltd is the only landfill mining company which can boast of an experience in clearing 150 acres of land from landfills and processing 25 lakh MT of legacy waste. The company till date has supplied more than 2.5 lakh MT of Refuse Derived Fuel, equivalent to more than 90,000 MT of CO₂e, to the cement companies which it uses as alternate fuel to coal, a non-renewable fossil fuel.

ASSOCIATIONS FOR GROWTH

Zigma has a technical association with South Korea based landfill Mining major Forcebel Co. Ltd. for understanding the best practices in the space and machinery designing. The company is credited with reclaiming more than 100 dumpsites across the globe. Some of the notable projects are the Seoul World Cup Stadium, Suwon Subway Station, Fukushima Radiation site and Japanese Tsunami Reclamation.

Zigma also saw a significant investment from Blue Planet Environmental Solutions Pte Ltd (Blue Planet), Asia's leading sustainable waste management company. The Singapore headquartered company driving regional sustainability through technology-driven and IP based end-to-end solutions for waste management and upcycling

NAVIGATING THE PANDEMIC

Since waste management services come under Essential Services Management Act, Zigma had little challenges in the sector.

HOWEVER, THE FOLLOWING PRECAUTIONS WERE UNDERTAKEN TO ENSURE THAT THE OPERATIONS WENT SMOOTH:

- *Employees working were provided accommodation separately to ensure complete isolation from the general public.*
- *Employees were provided with all precautionary medicine like vitamins,*

hot water and home remedies recommended by the Ayush Ministry to ensure that the virus spread was contained.

- *Employees were provided separate food.*
- *All employees worked with social distancing norms to ensure that the regulations put forward by the government were adhered to completely.*
- *All employees were provided transport facilities to ensure easy movement during lockdowns imposed.*

Working through the pandemic while taking utmost precautions, Zigma Global ensured that their altruistic motto to manage landfill waste in the country went on uninterrupted.

AWARDS AND ACHIEVEMENTS

Zigma Global has been showered with recognition and awards for its much-needed services to the country and the environment:

- *Featured under the Best Practises under the Solid Waste Management by the Swachh Bharat Mission in the year 2016*
- *Frost & Sullivan – Best Practices Award 2016 for Sustainable Waste Management*
- *Indian Leadership Award For Industrial Development from All India Achievers Foundation in the year 2017*
- *Award for 'Excellence' from Municipalika during 2017 for "Best Municipal Solutions"*
- *The company's NOIDA project was awarded the best Smart City Project, 2019*
- *The company's Vijayawada Project was awarded with the Mayor Puraskar, 2019*
- *The company's project in Tirupati featured in the Swachh Bharat Mission Advisory for Landfill Reclamation drafted by the Ministry of Housing and Urban Affairs (MoHUA), Government of India in 2020*
- *The company got featured in the Book of Innovations at the INK@WASH event organized by the Government of Telangana in the Municipal Solid Waste solutions for the year 2020*

Zigma Global Environ Solutions has till 2019 processed 2.5 million tonnes of legacy waste, but it aims to process 2.5 million tonnes of legacy waste this year alone. The vision is to see a landfill free nation in the days to come.



“Zigmaites are guided by the passion to converge technology and knowledge together to empower solutions and embark upon a methodology which can resolve the waste menace constantly harassing the Urban/Local Bodies and the surrounding communities at large. Our ethos is built on the principle of valuing the 'values' and trust skill over shortcuts.”

UNIQUE SELLING PROPOSITIONS

The only company which provides a near zero residue methodology in landfill mining space, Zigma provides a dedicated ERP system to the ULB's which help maintain complete transparency in the operation with live data feed. Zigma provides complete data points in terms of traceability on where and how the aggregates are disposed ensuring complete responsibility.

The company works on customization of machines which are self-built and are changed based on the characterization of